

Name: Anuttar Jain
Roll No: 22B00003

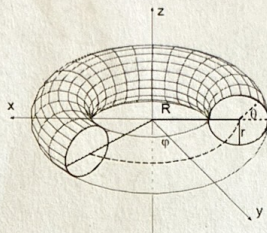
Indian Institute of Technology Bombay

PH 544 - Short Quiz 1

General relativity

Date 19-01-2026

1. Consider a donut (torus). The 2D metric that describes the surface of the donut is given by: $g_{\theta\theta} = (R + r \cos \phi)^2$, $g_{\phi\phi} = r^2$, and $g_{\theta\phi} = g_{\phi\theta} = 0$. r is the radius of the cross-section of the torus.



- (a) Compute the Christoffel symbols.

$$\Gamma_{ij}^k = \frac{1}{2} g^{lk} \left(\frac{\partial g_{il}}{\partial q^j} + \frac{\partial g_{jl}}{\partial q^i} - \frac{\partial g_{ij}}{\partial q^l} \right) \quad (1)$$

- (b) Using the Euler-Lagrange equation, obtain the equations of motion for the particle constrained on the torus

$$\ddot{q}^i + \Gamma_{jk}^i \dot{q}^j \dot{q}^k = 0 \quad (2)$$

$$(a) [g_{ij}] = \begin{bmatrix} (R + r \cos \phi)^2 & 0 \\ 0 & r^2 \end{bmatrix} \quad [g^{ij}] = \begin{bmatrix} \frac{1}{(R + r \cos \phi)^2} & 0 \\ 0 & \frac{1}{r^2} \end{bmatrix}$$

$$\Gamma_{\theta\theta}^{\theta} = \frac{1}{2} (\{0 + 0\} + \{0 + 0\} - \{0 + 0\}) = 0$$

$$\Gamma_{\theta\phi}^{\theta} = \frac{1}{2} \left\{ (2(R + r \cos \phi)(-r \sin \phi) \frac{1}{(R + r \cos \phi)^2} + 0) + (0 + 0) - 0 \right\}$$

$$\rightarrow = \frac{-r \sin \phi}{R + r \cos \phi}$$

$$\Gamma_{\phi\theta}^{\theta} = -\frac{r \sin \phi}{R + r \cos \phi}$$

$$\Gamma_{\phi\phi}^{\theta} = 0 \quad ; \quad \Gamma_{\theta\theta}^{\phi} = 0 \quad ; \quad \Gamma_{\theta\phi}^{\phi} = \Gamma_{\phi\theta}^{\phi} = 0$$